PRASAD MALWADKAR

Seattle, WA – malwadkarprasad99@gmail.com – linkedin.com/in/prasad-malwadkar – +1(213) 272-5083

SUMMARY

Senior Software Engineer with 3+ years' experience building backend, cloud systems & LLM-powered applications

EDUCATION

University of Southern California, **Master of Data Science** Jan 2024 - Dec 2025

Coursework: Machine Learning, Predictive Analytics, Algorithms, Data Mining, Advanced Data Stores, AI

EXPERIENCE

Flair Labs (YC W24)– **Founding Backend Engineer** April 2026-Present

- Spearheaded the technical design of a multi-tenant voice architecture using Redis distributed locks, Cloud Run, and GCP, scaling high-concurrency enterprise traffic and driving a **30% increase in lead conversion rates**.
- Directed a core **infrastructure** overhaul **and seamless provider migration**, leveraging **distributed systems** to **cut cloud egress costs by 40%** boosting platform resilience for **millions of daily outbound interactions**.
- Built **high-concurrency speech-to-speech** number rotation engine scaling **40k–80k calls/day** to safeguard reputation and enforce DNC compliance.
- Directed technical execution and architecture reviews for major strategic accounts orchestrating **live transfer routing and driving complex platforms from zero to production deployment**.

Amazon – **SDE Intern** May 2025 - Aug 2025

- **Led the end-to-end automation** of secure key rotation for **EC2 instances** using **cloud-formation** to expose standardized infra tools to AI agents and reducing manual ops overhead **by 80%**
- **Owned** migration of legacy test infrastructure to a microservices-based architecture, enabling AI-driven validations via CI/CD pipelines that **improved system reliability by 30%**

Amdocs – **Senior Software Developer** May 2023 - Dec 2023

- Architected and optimized **enterprise data pipelines** using Talend and Informatica, ingesting complex GraphQL datasets into HDFS while implementing OAuth 2.0 security and improving API response latency by **30%**.
- Spearheaded complex **ETL job tuning and query optimization** across large-scale Hadoop and HBase clusters, leveraging **Hive** and **PySpark** to accelerate data transfer rates by **40%** and streamline analytical workflows.
- Designed backend predictive processing engines and integrated modern UI query layers, driving core data automation that significantly reduced manual processing overhead and subscriber churn.

Amdocs – **Software Engineer** Aug 2021 - May 2023

- Executed comprehensive **ETL and job tuning** initiatives on Informatica workflows, using critical path analysis to parallelize system bottlenecks and **reduce execution runtime by 45%**.
- Led a robust '**Three Gates' migration strategy** for **2.5M enterprise customers' data**, achieving 100% data consistency and maintaining 99.9% service uptime during high-volume production transitions.
- Developed scalable **event-driven batch processing architectures**, improving system maintainability and lowering processing error rates across core data layers.

PROJECTS

Real-Time AI Voice Orchestrator - FastAPI, Gemini, Cartesia, Playwright, Twilio, STT, TTS

- Architected a first-principles **voice gateway** utilizing a custom **WebSocket bridge** to orchestrate low-latency audio streams between Twilio and Gemini, achieving sub-second Time to First Audio
- **Engineered** a state-management engine with **browser synchronization** using **pub/sub worker queues**.

Bloom Filter-Aided Hash Join Optimization - *C++, DuckDB, Data Stores, Query Optimization*

- Integrated Bloom filter pushdown directly into the DuckDB **query planner** in C++, reducing unnecessary CPU lookups by 45% while balancing **memory-CPU tradeoffs**.
- Achieved up to **2.5x speedup** on selective join workloads by optimizing query plan execution and filtering.

Fintech Agent Orchestrator - *FastAPI, React Flow, Next JS, SQLite*

- Architected a **decoupled multi-agent orchestration engine** using a custom state machine and a **Chunked Summary Buffer Memory**, cutting API token costs by **60%** while enabling dynamic real-time **agent handoffs** and zero-latency memory.

SKILLS

Languages & Databases	Python, TypeScript, JavaScript, Go, C++, SQL(Postgres), DynamoDB, Vector DB
Web Technologies	React, Next.js, Node.js, Django, RESTful APIs, GraphQL, Express, Redis
AI & Automation	LLM Integration, MCP Servers, LangChain, Predictive Analytics, RAG, MCP Servers
DevOps & Tools	AWS (Lambda, S3, EC2), Docker, Git, CI/CD, GitHub, Figma, AI workflows